

Exponential Expressions

MA.7.NSO.1.1

I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

Yesterday: seven laws, one at a time.

Today: several laws in one expression — the order is yours.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $\frac{10^5}{10^3}$
2. Is $\sqrt{50}$ closer to 7 or to 8?
3. Solve $4(x - 2) = 12$
4. Evaluate $2 + 3 \cdot 4^2$

Warm-Up

Answers.

1. Find the value: $\frac{10^5}{10^3}$ **100**
2. Is $\sqrt{50}$ closer to 7 or to 8? **7**
3. Solve $4(x - 2) = 12$ **$x = 5$**
4. Evaluate $2 + 3 \cdot 4^2$ **50**

Notes

Copy the table. It is your reference for the rest of the unit.

I. The seven laws, in one place

Law	Rule	Example
Product of powers	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^1 = 2^4$
Quotient of powers	$a^m \div a^n = a^{m-n}$	$(-3)^7 \div (-3)^2 = (-3)^5$
Power of a product	$(ab)^m = a^m \cdot b^m$	$(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$
Power of a quotient	$(a \div b)^m = a^m \div b^m$	$(13 \div 5)^3 = 13^3 \div 5^3$
Power of a power	$(a^m)^n = a^{mn}$	$(6^3)^4 = 6^{12}$
Identity exponent	$a^1 = a$	$15^1 = 15$
Zero exponent	$a^0 = 1$	$127^0 = 1$

Notes

Two students, two orders, one value.

II. More than one law — make a plan, then follow it

$$\text{Winston: } \left(\frac{2^6}{2^4} \right)^3 = (2^2)^3 = 2^6 = 64$$

$$\text{Yvette: } \left(\frac{2^6}{2^4} \right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^6 = 64$$

Different plans, same value. Any correct sequence of laws gives the same answer.

Notes

III. Order of operations still applies

- A. Parentheses first — including a fraction bar, which is a pair of parentheses in disguise.
- B. Exponents next — the laws of exponents are applied here.
- C. Then multiply and divide, left to right; then add and subtract.
- D. **Never across a + or a -:** $2^3 + 2^2 = 8 + 4 = 12$, **not** 2^5 .

Example 1

A zero exponent inside a product inside a quotient.

$$\frac{4^5 \cdot 4^0}{4^2}$$

Make a plan. Then find the value.

Example 1

Worked — Caroline's plan and Jamir's plan.

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^{5+0}}{4^2} = \frac{4^5}{4^2} \quad \text{Caroline: product of powers first}$$

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^5 \cdot 1}{4^2} = \frac{4^5}{4^2} \quad \text{Jamir: zero exponent first}$$

$$\frac{4^5}{4^2} = 4^3 = 64 \quad \text{both: quotient of powers, then evaluate}$$

Answer: 64

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{5^4 \cdot 5^0}{5^2}$$

Your Turn

Answer.

$$\frac{5^4 \cdot 5^0}{5^2}$$

$5^4 \cdot 1$ over 5^2 , so 5^2

Answer: $5^2 = 25$

Example 2

Three laws, a decimal and a negative base.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

Make a plan. Then find the value.

Example 2

Worked.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4} = 1 \cdot \left(\frac{-15}{1.5}\right)^4 \quad \text{zero exponent; power of a quotient, backwards}$$

$$\left(\frac{-15}{1.5}\right)^4 = (-10)^4 \quad \text{divide inside: } -15 \div 1.5 = -10$$

$$(-10)^4 = 10000 \quad \text{four negative factors: positive}$$

Answer: 10,000

Your Turn

Same steps, your numbers. Boards up when done.

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

Your Turn

Answer.

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

$$1 \cdot (-12 \div 4)^3 = (-3)^3$$

Answer: -27

Whiteboards • Question 1 of 9

Boards up on three.

A warehouse holds 6^4 boxes in 6^2 equal rows.

Each box weighs 3 kilograms.

How many boxes are in each row?

Answer it.

Whiteboards • Question 1 of 9

Answer.

A warehouse holds 6^4 boxes in 6^2 equal rows.

Each box weighs 3 kilograms.

How many boxes are in each row?

6^4 over 6^2 , so 6^2

Answer: 36

Whiteboards • Question 2 of 9

Boards up on three.

$$(3^0 \cdot 3^2)^3$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$(3^0 \cdot 3^2)^3$$

$$(3^2)^3 = 3^6$$

Answer: 729

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(0.5^3)^2}{(0.5)^2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(0.5^3)^2}{(0.5)^2}$$

$$0.5^6 \div 0.5^2 = 0.5^4$$

Answer: 0.0625

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

Answer it.

Find the value.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{2.5^5}{2.5^3} \right)^1$$

2.5^2 — the exponent 1 changes nothing

Answer: 6.25

Whiteboards • Question 5 of 9

Boards up on three.

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. $1/4$

B. 2^{22}

C. 0

D. 4

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. $1/4$

B. 2^{22}

C. 0

D. 4

Answer: D — 4

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. $0.2^0 \cdot 0.1^2$

B. $0.2^3 \cdot 0.1^2$

C. 0.1^5

D. 0.1^6

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. $0.2^0 \cdot 0.1^2$

B. $0.2^3 \cdot 0.1^2$

C. 0.1^5

D. 0.1^6

Answer: C — 0.1^5

Whiteboards • Question 7 of 9

Boards up on three.

$$0.1^2 \cdot (0.1^1)^2$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$0.1^2 \cdot (0.1^1)^2$$

$$0.1^2 \cdot 0.1^2 = 0.1^4$$

Answer: 0.0001

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

$(-4)^3$ over 1

Answer: -64

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

Write your answer in sentences.

Name the law each one used first, and give the value.

Whiteboards • Question 9 of 9

Answer.

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

both plans end at 2^6

Answer: Yes — both give $2^6 = 64$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate expressions using properties of exponents — ZE9